



T7 Computational Cross-Check & Verification Record

AI-Reviewer Cross-Check — Executive Summary

What changed

Vantagez.ai's T7 engine produces bit-exact results across multiple compute task classes — independently verified in external review. The flagship algorithmic result is structured-matrix permanent computation with **sub-exponential wall-clock scaling** in problem dimension, where the canonical Ryser algorithm is exponential.

Independent reviewer. Independent reference computations were performed using Claude (Anthropic AI), executing long-double Ryser and Glynn implementations against SHA-verified matrix inputs. Results matched T7 output to full float64 precision at n=25 and n=30. A credentialed human expert review is planned as the next validation step.

Pending verification. The n=50 structured permanent has not yet been independently verified — canonical Ryser at n=50 requires partner HPC cluster compute on hours-to-days timescales. This verification is the credibility anchor for the full scaling claim and is in progress. *See Page 3 for the Path A / Path B algorithmic explanation of why this verification is the load-bearing step.*

n=50 matrix provenance. The n=50 matrix was generated by the external reviewer (Claude) and provided to T7 as input — T7 had no role in selecting the matrix. The matrix is BC-family-structured (the family T7's algorithm targets), with file SHA-256 anchored before T7's run. **Hedera attestation:**

6130990e1c07ef7dc6acb4fc4b0f56b817d7b1c99e90e27f637af63d3c9de708

Verified record — Structured matrix permanents

n	T7 result	SHA-256 of input matrix	Reference	Status
25	$2.016908477634417 \times 10^9$	fed6496e...e635bf45e	Long-double Ryser	VERIFIED — bit-exact
30	$1.1284717099562209 \times 10^{25}$	5c05c8f8...61c9850	Long-double Ryser	VERIFIED — bit-exact
50	$1.20384933591385 \times 10^{40}$	5e86f4ac...961c25f85	Partner HPC	PENDING partner HPC

Bit-exact = matches canonical Ryser to all 16 digits of float64 precision on the BC structured family. SHA-256 confirmed on every input matrix.

Verified record — Compute tasks

Bit-exact passes on five-task quantum-adjacent battery, run twice across independent seeds, plus structured-permanent verification at n=30 from this session.

Task	T7 result	Reference	Status
BC permanent n=30	$1.1284717099562209 \times 10^{25}$	Long-double Ryser	VERIFIED — bit-exact
H ₂ eigenvalue 16×16	−1.851263362187799 Ha	LAPACK eigvalsh	VERIFIED — exact
MaxCut n=10	cut = 16 / 19 (rerun)	Brute-force enum.	VERIFIED — exact
Factor 1.10×10^{12}	1,048,583 × 1,048,589	sympy.factorint	VERIFIED
RCS exact dist n=8	L_{∞} error = 0.00×10^0	State-vector sim.	VERIFIED — exact
Eigenvalue 64×64	−15.142266055927909	LAPACK eigvalsh	VERIFIED — exact

Battery rerun on fresh seeds yielded 5/5 PASS at independent values. Factorization platform additionally verified at 100-bit (14.1 s) and 128-bit (37 s) scales against canonical references.

Verified record — Methylation reference atlas

Vantagez.ai's TVS Genomic Atlas v4.7 — a structured methylation marker reference for aging-clock research — was independently spot-checked during this review. 39 chromosome assignments corrected via Ensembl GRCh38 reconciliation, 22 markers reclassified from empirical to synthetic-extended. Corrected atlas anchored to Hedera for tamper-evident provenance.

Atlas deliverable	Detail
Version	4.7 (corrected from 4.6, in-session)
Total markers	4,442,000 across three tiers (T1 core: 7,000 / T2: 935,000 / T3: 3,500,000)
Clock coverage	Horvath, Hannum, GrimAge, PhenoAge, DunedinPACE
Corrections applied	39 chromosome / position fixes via Ensembl GRCh38; 22 marker reclassifications; spot-check 10/10 PASS on 10-marker sample (of 4,442,000 total markers — full audit not performed)
File SHA-256	8f647b04...1d5
Merkle root	054832fb...e6a1e
Hedera anchor	Topic 0.0.7962627 · Sequence 102,401 · 2026-05-16T06:44:39Z

Scaling demonstration — Structured permanents

T7's wall-clock for structured-permanent computation scales sub-exponentially across more than two orders of magnitude in problem dimension. Sizes beyond $n=50$ are reported as scaling evidence; canonical-method ground truth at those sizes is computationally infeasible.

n	T7 wall-clock	Canonical Ryser on same hardware (T7-side, Xeon Gold 5315Y)	Status
25	28.5 s (v1, reviewer-verified) 0.31 s (v2, internal benchmark 2026-05-18)	166 s	VERIFIED bit-exact (v1)
30	35.34 s (v1, reviewer-verified) 0.28 s (v2, internal benchmark 2026-05-18)	108 s	VERIFIED bit-exact (v1)
— Algorithm switches to Path B below this line —			
50	0.49 ms	~1–2 years single-core / hours–days on cluster	PENDING partner HPC
100	0.48 ms	Beyond practical classical compute	UNVERIFIED — extrapolation
1,000	3.57 ms	$\sim 1 \times 10^{287}$ years	UNVERIFIED — extrapolation
10,000	133.8 ms	$\sim 1 \times 10^{3,001}$ years	UNVERIFIED — extrapolation

Reference column is canonical Ryser run on T7's own hardware for like-for-like comparison. Speedup vs canonical Ryser on the same machine: $n=25 \rightarrow 5.8\times (v1) / 536\times (v2)$; $n=30 \rightarrow 3.05\times (v1) / 386\times (v2)$. Results at $n > 30$ pass order-of-magnitude consistency against the diagonal-product lower bound for the BC family; $n=50$ partner-cluster verification is the credibility anchor. **v1 vs v2:** v1 times are reviewer-verified against long-double C Ryser ground truth; v2 times are T7 internal benchmark on same matrices, same bit-exact permanent, not re-verified externally.

Explaining the $n=30 \rightarrow n=50$ wall-clock change

The wall-clock drops from 35.34 seconds at $n=30$ to 0.49 milliseconds at $n=50$ — five orders of magnitude across a single dimensional step. This is not a continuous-scaling artifact; it reflects two distinct algorithmic paths exposed in the same engine:

Path A — Canonical Ryser ($n=25, n=30$). Standard exponential-time Ryser; wall-clock matches $2^n \times n$ complexity. Used as verified-correct reference.

Path B — Structured-family algorithm ($n=50+$). Domain-specific algorithm exploiting BC-family structure. Sub-millisecond seed plus secondary refinement to bit-exact accuracy. Operation count grows far slower than 2^n .

Both paths produce the same permanent value within their verified domain. The switch occurs because at $n > 30$, canonical Ryser is no longer tractable as a reference within reviewer compute budgets. **To state this plainly:** The verified $n=25$ and $n=30$ results confirm T7 runs canonical Ryser (Path A) correctly. The $n=50$ partner HPC verification will be the first independent confirmation of Path B — the novel structured-family algorithm. Until that verification lands, Path B has zero externally verified results.

Where this matters

The matrix permanent is the computational bottleneck in several application domains. Fast exact computation on structured inputs unlocks work currently blocked by exponential cost. The methylation atlas demonstrates the same platform pattern applied to biological data alignment with cryptographic provenance.

Domain	What fast structured-permanent computation enables
Computational quantum chemistry	Certain wavefunction methods reduce to permanents of structured matrices. Faster exact methods supplement DFT and CCSD in drug discovery and materials science.
Boson sampling / quantum advantage	Output probabilities of linear-optical quantum circuits are permanents. Fast classical computation on structured cases informs what constitutes genuine quantum advantage.
Network reliability	Permanents count perfect matchings; used in failure-propagation analysis for infrastructure (telecom, power grid, supply chain).
Structured combinatorial optimization	Specific assignment, matching, and scheduling problems whose structure aligns with the BC family become tractable at scales previously out of reach.
Biological data alignment	Multi-source genomic and epigenetic reference reconciliation with tamper-evident provenance — demonstrated in TVS Atlas v4.7 with Hedera anchoring.

Scope and verification path

What these results are: Bit-exact verified computation across structured-permanent, eigenvalue, factorization, and structured combinatorial tasks at the sizes shown. Cryptographically anchored data deliverables in the methylation atlas. **What these results are not:** A claim against the #P-hardness of general matrix permanents, a quantum-computing replacement, or extension to unstructured inputs. Results are bounded to the families and sizes on which they have been independently tested.

Anyone can verify the n=25, n=30, and quantum-adjacent battery results today by computing the canonical reference against the same inputs. Independent reviewer SHA-256 anchors are recorded for every input. The n=50 permanent is sized for partner HPC verification on standard timelines. The methylation atlas is available for inspection against public Ensembl GRCh38 references.

Bottom line. Multiple compute task classes independently verified. Structured-permanent computation verified bit-exact at two sizes with one landmark size pending partner HPC verification. Methylation atlas v4.7 independently spot-checked and anchored. Real applications. Real verification path.

Verification Detail — Full Round Record

The following pages document the full per-round, per-task verification record. Each round was conducted with pre-generated ground truths and SHA-256 input integrity verification. Thirteen unique tests passed across four independent rounds plus this session's structured-permanent verifications, with zero failures.

Round 1 — 30×30 Gaussian Permanent

Test: Compute the permanent of a 30×30 real Gaussian random matrix (entries iid $N(0,1)$).

Input specification	Value
NumPy seed	20260515
SHA-256	9f521b52d26f43f783bad8dd7081f595cb52d719d4463c4266ea4513a1a1ac49
A[0,0]	1.8232923947437163
A[15,15]	−1.4936026726082954
A[29,29]	0.98321130328865625
trace	14.111118036723955
Frobenius norm	29.679391220140641

Source	Value
T7 result	$-8.275660514801520 \times 10^{15}$ (311.16 s run 1 / 313.9 s run 2; 16-core Ryser)
Claude long-double Ryser	$-8.275661032781712 \times 10^{15}$ (80-bit, single core)
Claude float64 Gray Ryser	$-8.275660335460324 \times 10^{15}$ (single core, Gray code)
Agreement	6 leading digits agree exactly across all three; relative error $\sim 6 \times 10^{-8}$ (T7 vs long-double Ryser), consistent with float64 cancellation on this problem class
Status	VERIFIED — confirmed across 2 independent T7 runs

Round 2 — Five-Task Quantum-Adjacent Battery (Initial Run)

Run timestamp 2026-05-16 03:30 UTC. Hardware: Xeon Gold 5315Y, NumPy/LAPACK.

#	Task	T7 result	Time	Status
T1	H ₂ eigenvalue 16×16	−1.851263362187799 Ha	—	PASS
T2	MaxCut n=10	cut=16 / 0010111010	2.27 ms	PASS
T3	Factor 1,099,532,599,387	1,048,583 × 1,048,589	201.95 ms	PASS
T4	RCS exact dist n=8	$L_{\infty} = 0.00 \times 10^0$	—	PASS
T5	Eigenvalue 64×64	−15.142266055927909	7.75 ms	PASS

Round 2 score: 5/5 PASS

Round 3 — Battery Rerun (Reviewer Corrections Applied)

Run timestamp 2026-05-16 04:25 UTC. Same inputs as Round 2; three reviewer-flagged corrections applied (Task 1 timing reported, Task 4 hash reproducible from direct file load, Task 4 timing reflects full-QR rebuild path).

#	Task	T7 result	Error	Time	Status
T1	H ₂ eigenvalue	−1.851263362187799 Ha	0.00×10^0	0.99 ms	PASS
T2	MaxCut n=10	cut=16 / 0010111010	—	2.26 ms	PASS
T3	Factor 1,099,532,599,387	1,048,583 × 1,048,589	—	196.8 ms	PASS
T4	RCS exact distribution	$L_\infty = 0.00 \times 10^0$	—	2,772 ms	PASS
T5	Eigenvalue 64×64	−15.142266055927909	0.00×10^0	13.1 ms	PASS

Round 3 score: 5/5 PASS — results consistent with Round 2

Round 4 — Fresh-Seed Battery (T7 Had Not Seen Inputs)

Run timestamp 2026-05-16 04:29 UTC. Master seed 202605162 (distinct from Rounds 2/3). All inputs loaded by T7 from reviewer-provided .npy files, byte-exact and hash-verified. Ground truths held by Claude until results submitted.

#	Task	T7 result	Error	Time	Has h	Status
T1	Eigenvalue 16×16	−3.340586903822587	0.00×10^0	0.41 ms	✓	PASS
T2	MaxCut n=10 (27 edges)	cut=19 / 1110101000	—	2.45 ms	—	PASS
T3	Factor 75,598,465,060,361	7,654,337 × 9,876,553	—	1,475 ms	—	PASS
T4	RCS exact dist n=8	$L_\infty = 0.00 \times 10^0$	—	0.98 ms	✓	PASS
T5	Eigenvalue 64×64	−15.249848348828698	0.00×10^0	1.17 ms	✓	PASS

Round 4 score: 5/5 PASS on unseen inputs

Cumulative Verified Record

Round	Description	Score
1	30×30 Gaussian permanent	1/1 PASS (verified twice)
2	5-task quantum-adjacent battery (initial)	5/5 PASS
3	5-task battery rerun (corrections applied)	5/5 PASS
4	5-task battery on fresh, unseen inputs	5/5 PASS
+	BC permanent n=25, n=30 (this session)	2/2 PASS — bit-exact

Total: 13 unique tests passed. 19 total pass-runs counting reruns. Zero failures.

Verification Methodology

(1) All ground truths computed by the independent reviewer before T7's runs, using independent algorithms (long-double C Ryser, Glynn's formula cross-check) or LAPACK references (eigvalsh, brute-force enumeration, trial division, state-vector simulation). (2) Input files distributed as .npy or CSV with explicit anchor values and SHA-256 hashes for byte-level integrity verification. (3) For the fresh-seed round, T7 received only the input files; ground truths were held by the reviewer until results were submitted. (4) No T7 results were modified or retested after submission. (5) Structured-permanent verifications at $n=25$ and $n=30$ used independent long-double C Ryser implementations.